SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2014

Paper 1

H2 BIOLOGY
9648

Friday
29 August 2014

1 hour 15 minutes

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write you name, CG and index number in the spaces at the top of this page.

Enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C**, **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

AT THE END OF THE EXAMINATION, HAND IN BOTH OMS AND QUESTION PAPERS.

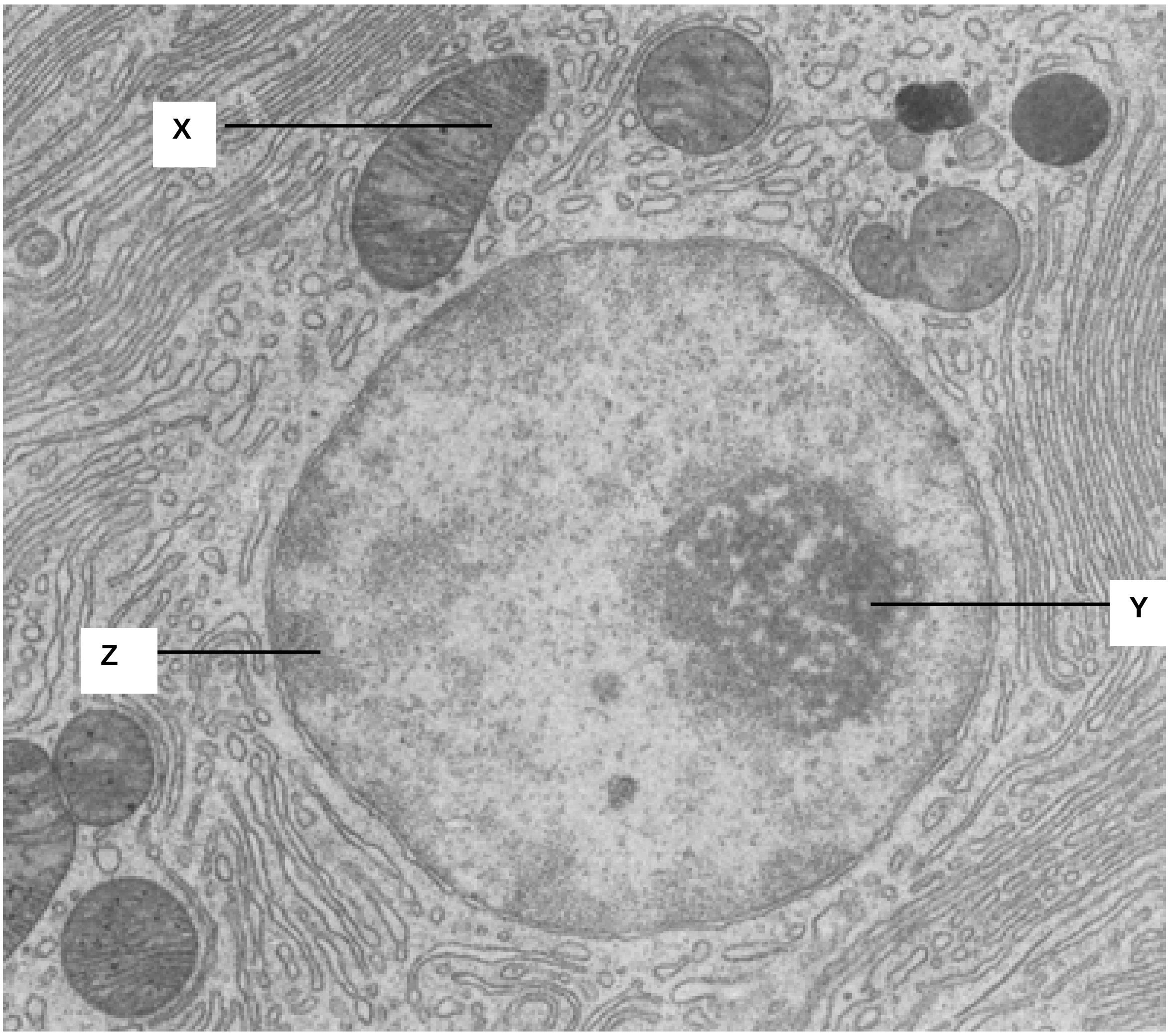
INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the question paper.

This question paper consists of 32 printed pages.

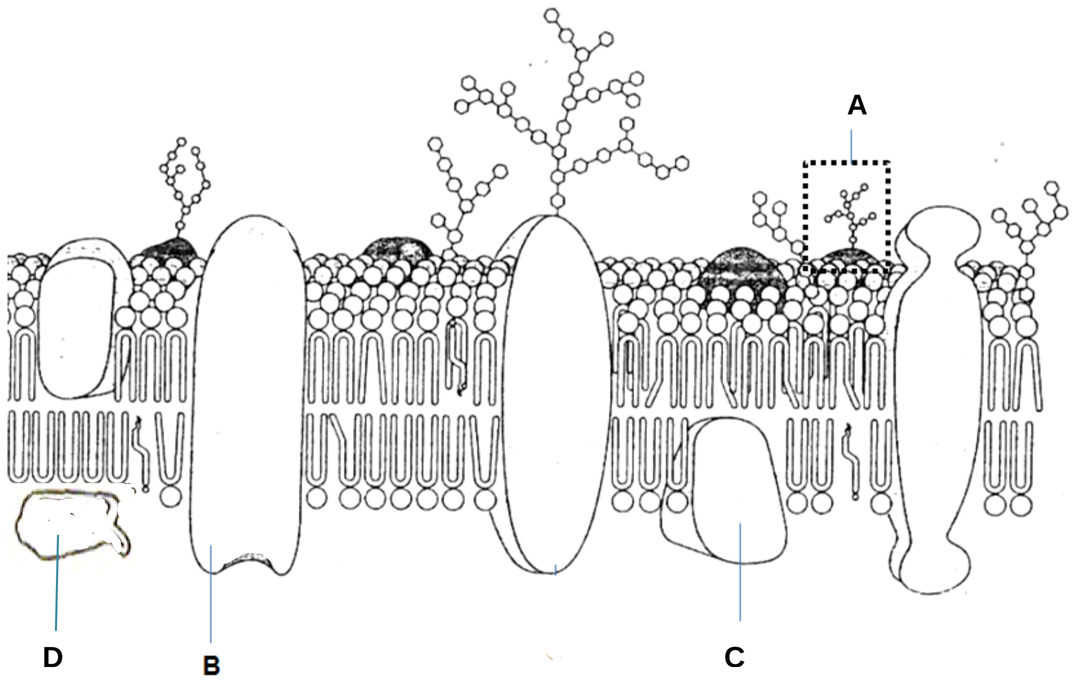
1. The electron micrograph below shows the structures found in a cell.



Which of the following statements is true for structures **X**, **Y** and **Z**?

	X	Y	Z
A	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD^+ and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Contains reduced NAD^+ and reduced FAD	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

2. The diagram below shows a section of a cell surface membrane.



Which of the following statements are correct?

I Structure **A** is found only on the extracellular surface of the membrane.

II Structure **B** may contain a channel that is hydrophobic to allow ions to move across the membrane.

III Structure **B** is able to bind to certain substances and change its conformation to transport these substances into the cell.

IV Structure **C** allows transport of substances across the membrane only in the presence of ATP.

V Structure **D** is found only on one surface of the membrane.

A **I, II, III** and **IV**

B **II, III** and **V**

C **I** and **III** only

D **I** and **V** only

3. A gland cell capable of producing large quantities of the sex hormone estrogen would be likely to have an abundance of

A lysosome

B ribosomes

C rough endoplasmic reticulum

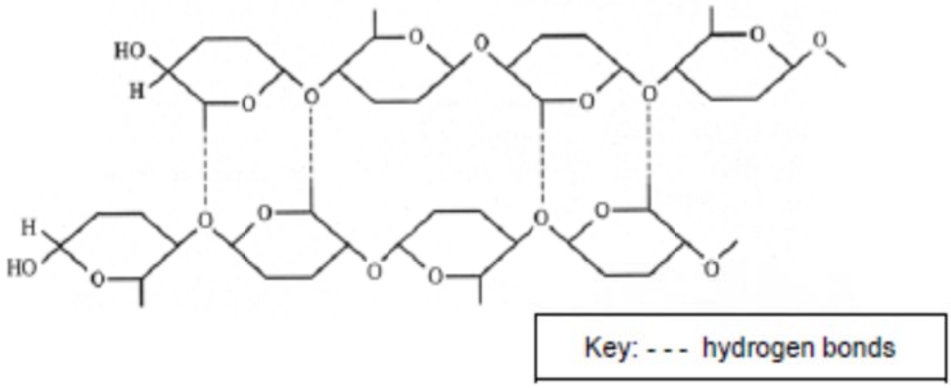
D smooth endoplasmic reticulum

4. The cell membrane of an African desert rat and that of the Arctic red fox are compared to investigate their composition. Which of the following is the most probable makeup of the respective membranes?

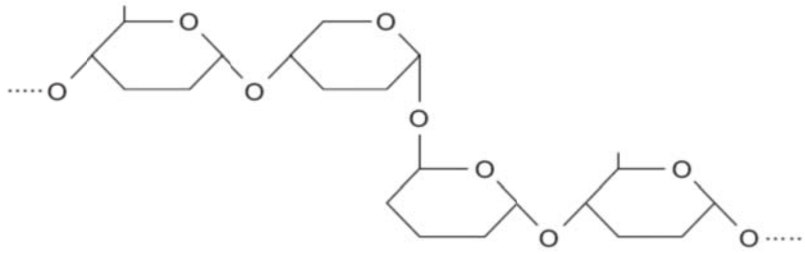
	Amount of saturated fatty acids		**Amount of cholesterol**	
	Arctic red fox	**Desert rat**	**Arctic red fox**	**Desert rat**
A	Low	High	Low	High
B	High	Low	Low	Low
C	Low	High	High	High
D	High	Low	High	Low

5. The figure below shows part of the molecular structure of two polymers, **A** and **B**.

Polymer A:



Polymer B:



Which of the following statement are true?

I **A** contains β-glucose while **B** contains α-glucose.

II **A** is an important food source for animals but cannot be digested by humans, due to presence of strong glycosidic bonds.

III **A** may associate in groups to form microfibrils, which are arranged in larger bundles to form macrofibrils.

IV **B** is a storage molecule which is held by many inter-chain hydrogen bonds.

A **I, II** and **III**

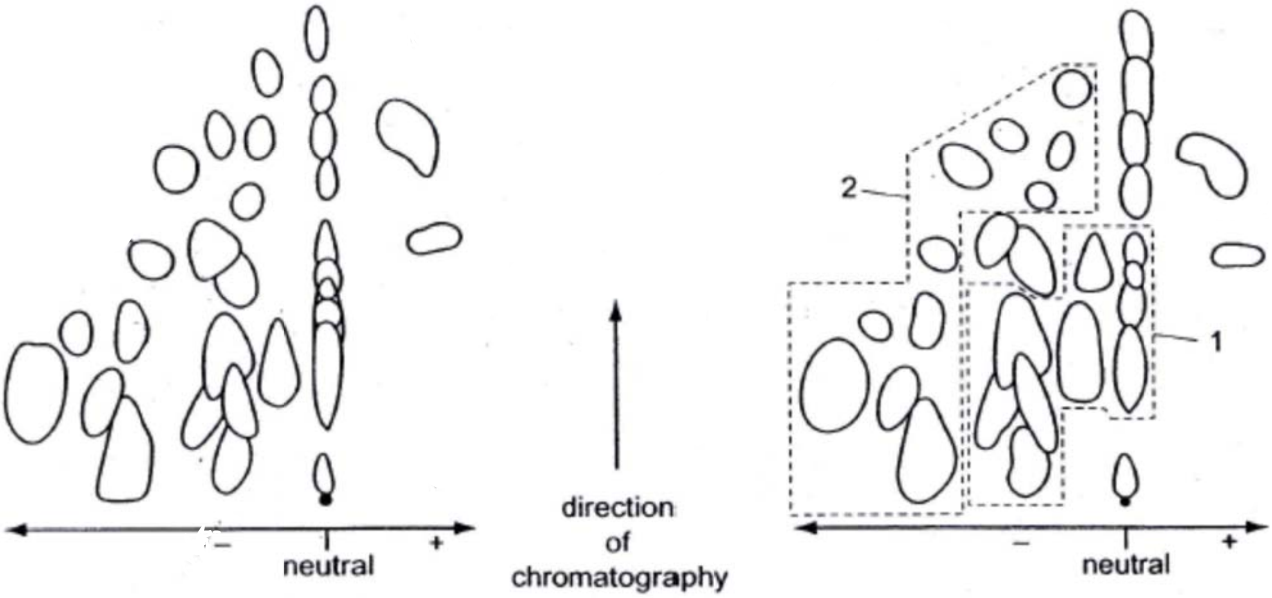
B **II** and **III**

C **I** and **III**

D All of the above

6. Early studies on sickle cell haemoglobin involved investigating the results of changed DNA nucleotide sequence. Haemoglobin was treated with a protease that hydrolysed it into several fragments. These were then separated by their charge using electrophoresis on a paper support. The paper was turned through 90^0 and chromatography was used to separate the mixture. The more soluble the amino acid, the further it is carried by the solvent as it moves in the direction of chromatography.

The diagrams show the results for normal and sickle cell haemoglobin.



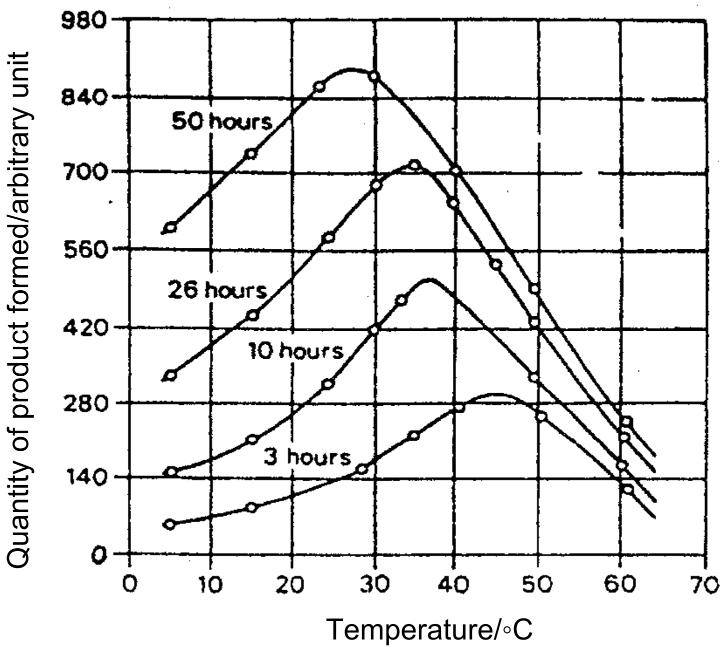
normal haemoglobin

sickle cell haemoglobin

Which describes the changes to haemoglobin and the fragments resulting from the changed DNA nucleotide sequence?

	Region of the diagram that shows the fragment containing the changed amino acid	Effect of changed DNA nucleotide sequence on amino acid sequence and charge of fragment
A	1	Glutamate (hydrophilic and polar) changed to valine (hydrophobic and non-polar), no longer balancing the charge of the rest of the fragment
B	1	Valine (hydrophobic and non-polar) changed to glutamate (hydrophilic and polar), more nearly balancing the charge of the rest of the fragment
C	2	Glutamate (hydrophilic and polar) changed to valine (hydrophobic and non-polar), no longer balancing the charge of the rest of the fragment
D	2	Valine (hydrophobic and non-polar) changed to glutamate (hydrophilic and polar), more nearly balancing the charge of the rest of the fragment

7. The graph below shows the quantity of the product formed when samples containing the same concentration of enzyme and substrate were kept at different temperature for four different durations.



Which statement best explains why the optimum temperature is lowered if the duration of incubation is increased?

A There is an increase in the denaturation of enzymes at high temperature.

B The activation energy of the reaction is lowered at high temperature.

C The product formed acts as an allosteric activator to enhance enzyme activity.

D More substrates are converted into products for longer durations of incubation.

8. In the production of isoleucine from threonine (**pathway P**), the end product acts as an inhibitor of the first enzyme in the pathway. In the production of arginine (**pathway Q**), the end product has no influence on other enzymes in the pathway. The two pathways are shown in the figure.

Pathway P:

inhibits enzyme 1

threonine $\xrightarrow[\text{enzyme 1}]{}$ compound X $\xrightarrow[\text{enzyme 2}]{}$ compound Y $\xrightarrow[\text{enzyme 3}]{}$ isoleucine

Pathway Q:

substrate $\xrightarrow[\text{enzyme 1}]{}$ compound M $\xrightarrow[\text{enzyme 2}]{}$ compound N $\xrightarrow[\text{enzyme 3}]{}$ arginine

Which of the following conclusions is correct?

I	In **pathway P**, if the production of enzyme 3 stops, there would be increased production of isoleucine.
II	In **pathway Q**, if the production of enzyme 3 stops, there would be decreased production of arginine.
III	In **pathway P**, provided all enzymes are present, the production of isoleucine would be continuous if there was a continuous supply of threonine.
IV	In **pathway Q**, provided all enzymes are present, the production of arginine would be continuous if there was a continuous supply of substrate.

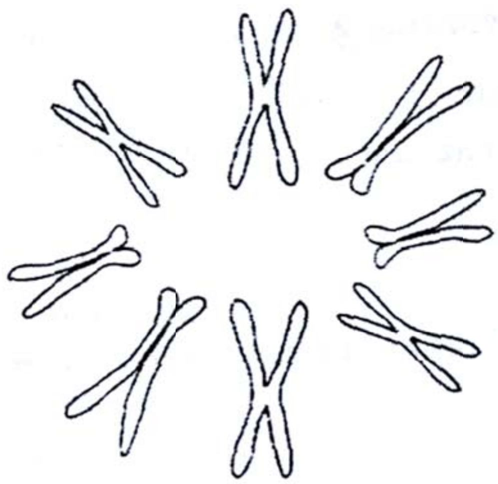
A **I** and **III** only

B **I** and **IV** only

C **II** and **III** only

D **II** and **IV** only

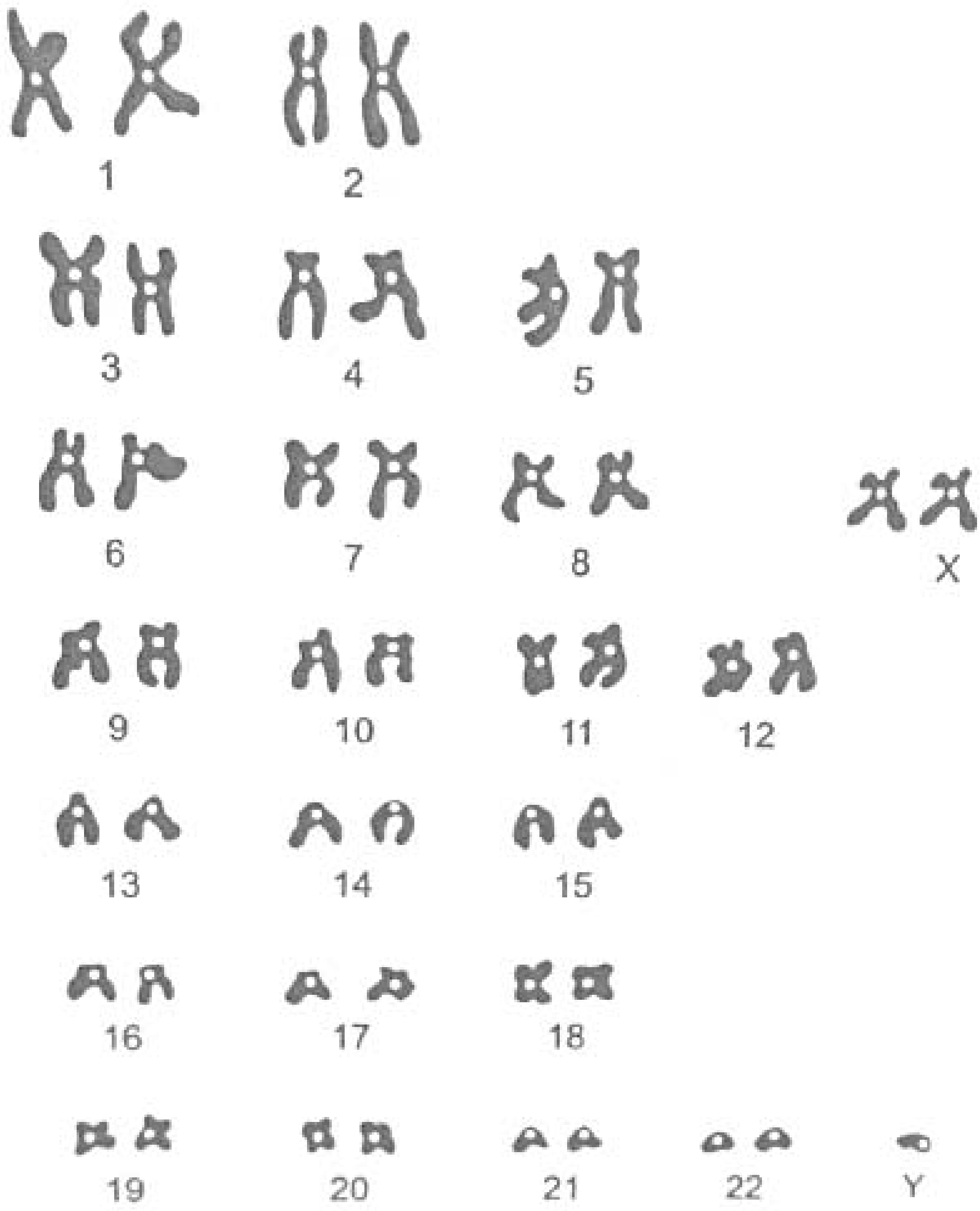
9. The diagram shows the behaviour of chromosomes in an animal cell during a particular stage of cell division.



Which of the following correctly describes the chromosomes shown?

	Stage of cell division	Number of chromosomes in a haploid set	Total number of DNA molecules
A	Metaphase	4	16
B	Interphase	4	16
C	Interphase	8	32
D	Metaphase	8	32

10. The figure below shows a human karyotype.



What can be concluded from the karyotype provided?

A There was non-disjunction during meiosis I in the mother.

B There was non-disjunction during meiosis II in the father.

C One contributory gamete to the zygote is an egg with an X and a Y chromosome.

D One contributory gamete to the zygote is a sperm containing two X chromosomes.

11. Two genes involved in coat color of goats are at loci on different chromosomes.

The color gene C causes the hairs to be uniform in colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^{B} giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^{G}) causes the development of white hairs between the coloured hairs, giving the coat a shaded appearance.

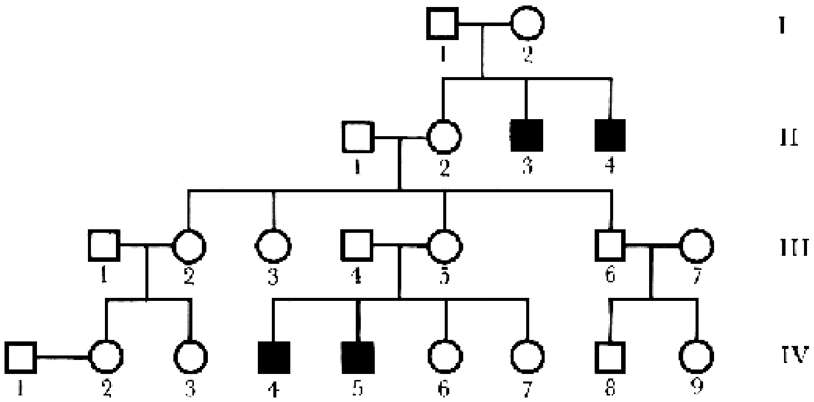
The table shows the results of crosses between a male goat and two female goats.

Cross	**Parents**	**Offspring**
1	Black agouti male **x** Uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
2	Black agouti male **x** Medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	Black agouti male	Uniformly dark brown female	Medium brown agouti female
A	$C^{B}C^{MB}A^{G}A^{G}$	$C^{B}C^{DB}A^{G}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$
B	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{B}C^{MB}A^{G}A^{G}$
C	$C^{B}C^{DB}A^{G}A^{G}$	$C^{DB}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{G}A^{g}$
D	$C^{B}C^{MB}A^{G}A^{g}$	$C^{B}C^{DB}A^{g}A^{g}$	$C^{MB}C^{MB}A^{G}A^{G}$

12. Haemophilia is a rare genetic disorder in which the blood does not clot normally. The mutation that causes haemophilia is located on the X-chromosome. The pedigree below shows the inheritance of haemophilia in a family. Squares represent males while circles represent females. Dark symbols represent affected individuals.



What is the probability that a son born to IV-2 will be a haemophiliac?

A 0.5

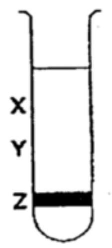
B 0.25

C 0.125

D 0.0625

13. A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N. The bacterial culture was then allowed to reproduce using nucleotides containing normal ^{14}N. The DNA was examined using a centrifuge after one generation and again after two generations.

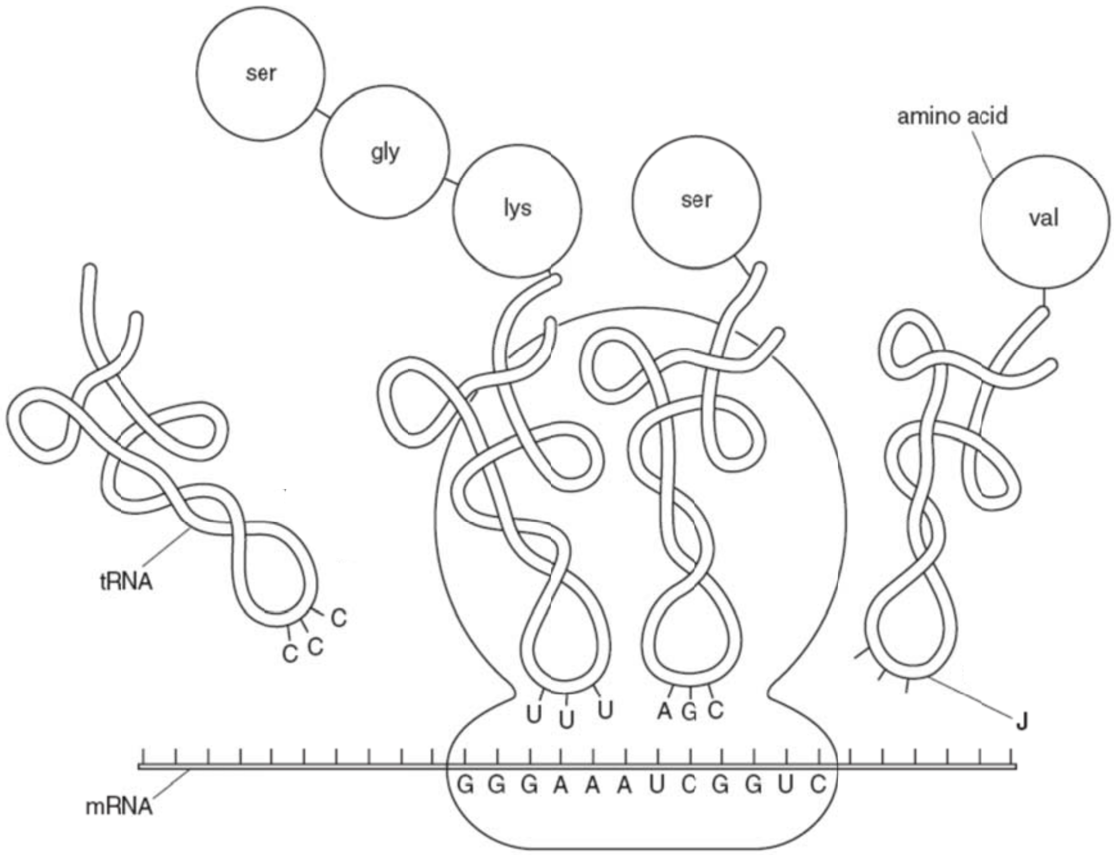
The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



Where would the DNA be found after the second and after the third cell generations?

	After second generation	After third generation
A	Half at X and half at Y	One quarter at X and three quarter at Y
B	Half at X and half at Y	One quarter at Y and three quarter at X
C	One quarter at X and three quarter at Y	Half at X and half at Y
D	One quarter at Y and three quarter at X	Half at X and half at Y

14. The diagram illustrates the process of translation.



Which of the following statements is / are true?

I	The base sequence at structure **J** is 3'–CCA–5'.
II	The anticodon of an incoming aminoacyl-tRNA base pairs with the complementary mRNA codon in the P site of the ribosome.
III	The movement of the ribosome is from the right to the left of the diagram above.
IV	A release factor causes the addition of a water molecule instead of an amino acid to the polypeptide chain, hydrolysing the completed polypeptide from the tRNA in the A site. This brings about the release of the polypeptide through the exit tunnel of the ribosome's large subunit.

A **IV** only

B **I** and **III** only

C **I, II** and **III** only

D None of the above

15. Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

first position	second position U	C	A	G	third position
U	phe phe leu leu	ser ser ser ser	tyr tyr STOP STOP	cys cys STOP trp	U C A G
C	leu leu leu leu	pro pro pro pro	his his gln gln	arg arg arg arg	U C A G
A	ile ile ile met	thr thr thr thr	asn asn lys lys	ser ser arg arg	U C A G
G	val val val val	ala ala ala ala	asp asp glu glu	gly gly gly gly	U C A G

Which event occurs to explain the information given above?

A There is no tRNA with an anticodon complementary to the first codon.

B The first codon is removed from the mRNA transcript in post-transcriptional modification.

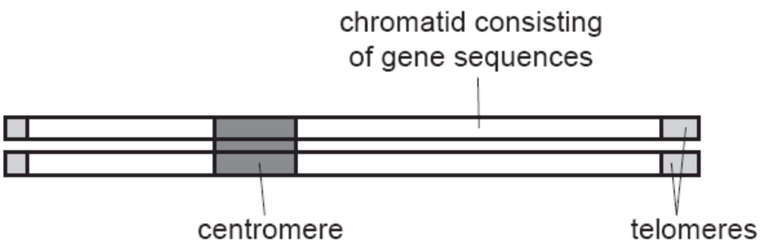
C The mRNA binds to the rRNA in the second codon position.

D The first amino acid on the polypeptide chain is removed in post-translational modification.

16. The ends of eukaryotic chromosome contain a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeric DNA.

The diagram shows a eukaryotic chromosome.



What is a consequence of the loss of repeating DNA sequences from the telomeres?

A The cell will begin the synthesis of different proteins.

B The cell will begin to differentiate as a result of the altered DNA.

C The number of mitotic divisions the cell can make will be limited.

D The production of mRNA will be reduced.

17. DNAse is added to samples taken from the same chromosomal region from three different cell types. After allowing for digestion of DNA, the remaining DNA samples are weighed with the following results

Cell type	Mass of DNA before adding DNAse/ µg	Mass of DNA after adding DNAse/ µg
A	5.0	5.0
B	5.0	3.4
C	5.0	4.1

What do these results suggest about genes and their expressions in all the three different cell types?

A Some regions of the cell type A chromosomes contain RNA.

B The genes found in the same chromosomal region in all the three cell types are expressed the most in cell type B.

C The mass of DNA in cell type A did not decrease because DNA is not of complementary shape to the active site of DNAse.

D There are more phosphodiester bonds in the DNA found in the chromosomes of cell type B than those in cell type A and C.

18. Which of the following mutations may directly lead to the occurrence of cancer?

1 germline mutation in a copy of p53 gene
2 somatic mutation in second copy of p53 gene
3 gene amplification of Ras gene
4 point mutation in a copy of Ras gene

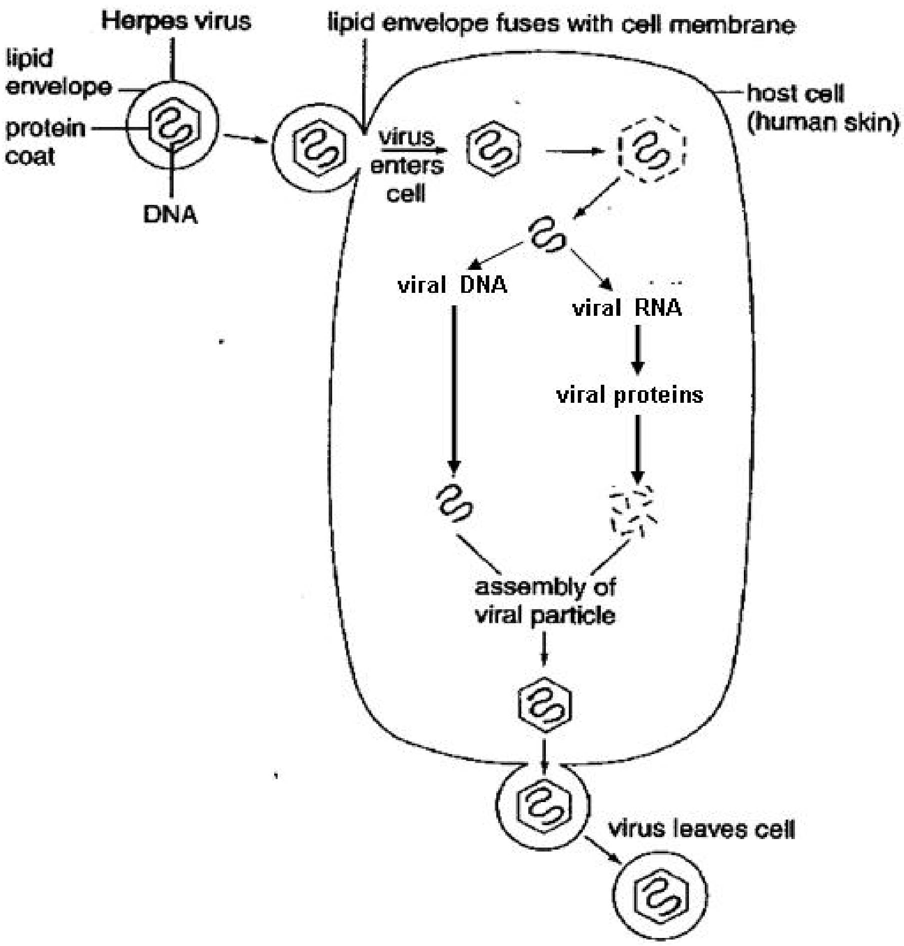
A 1 and 4

B 2 and 3

C 2, 3 and 4

D 1, 2, 3 and 4

19. The diagram below shows the reproductive cycle of the herpes virus which causes cold sores on the mouth. With reference to the diagram below, which of the following statements best describes the herpes virus?



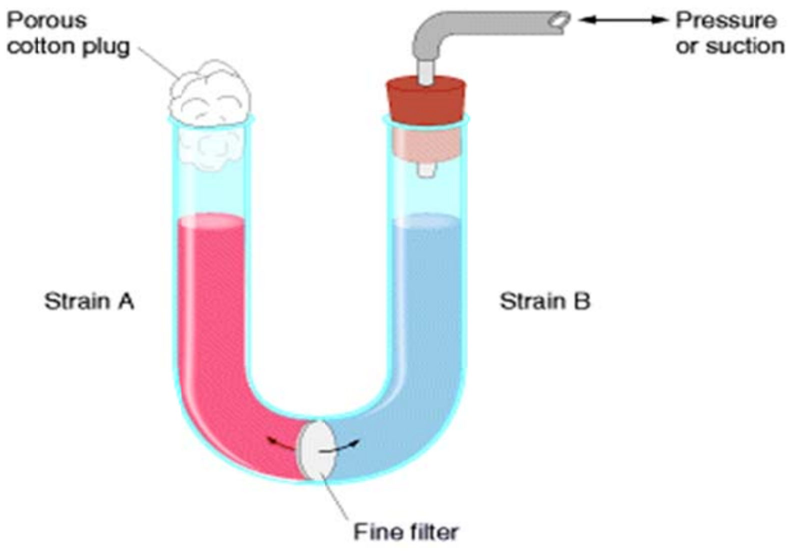
A It is not a retrovirus.

B Its mode of replication is similar to that of influenza virus.

C Its replication cycle includes a lysogenic phase.

D Death of the host cell is necessary for the release of the viral progeny.

20. To investigate gene transfer between bacteria, two strains of the same bacterial species were each placed in one arm of a U-tube with a filter separating them.



strain	genotype
A	*met* $^-$
B	*met* $^+$

met $^+$ is a wild-type gene that codes for the ability to synthesise the essential amino acid, methionine.

met $^-$ indicates that the *met* $^+$ gene has been mutated.

Liquid may be transferred between the arms of the tube by the application of pressure or suction, but particles that are larger than the filter pore size would not be able to pass through the fine filter.

type of particle	size
bacteria	1 – 10µm
bacteriophages	0.025 – 0.2µm

After several hours of incubation, bacterial cells from the left arm of the tube are plated on minimal medium.

Which pair of experimental results shows that transduction was most likely the process responsible for gene transfer between strains A and B?

	filter pore size	growth of colonies on minimal medium
A	5µm	no
	0.1µm	yes
B	5µm	no
	0.1µm	no
C	0.45µm	no
	0.02µm	no
D	0.45µm	yes
	0.02µm	no

21. A mutant strain of *E. coli* has been isolated in which the *lac* operon is not expressed in the presence of lactose. To understand the nature of this defect, this parent strain was mated so that it now contains an F' factor containing a normal *lac* operon. The parent and mated strain with regard to their beta-galactosidase activities in the presence and absence of lactose was compared. The following results were obtained:

Strain	**Addition of lactose**	**Amount of beta-galactosidase (percentage of parent strain)**
Parent	No	0
Parent	Yes	0
Mated	No	0
Mated	Yes	100

With respect to the results shown in table, which part of the operon most likely has the defect/mutation?

A *lac A*

B *lac I*

C Promoter

D Operator

22. Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

I A chemical that would bind to NAD^+ irreversibly and induce its reduction to NADH.

II An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.

III A chemical that creates an anaerobic environment by combusting in oxygen

IV A reagent that binds to the active site of ATPase permanently

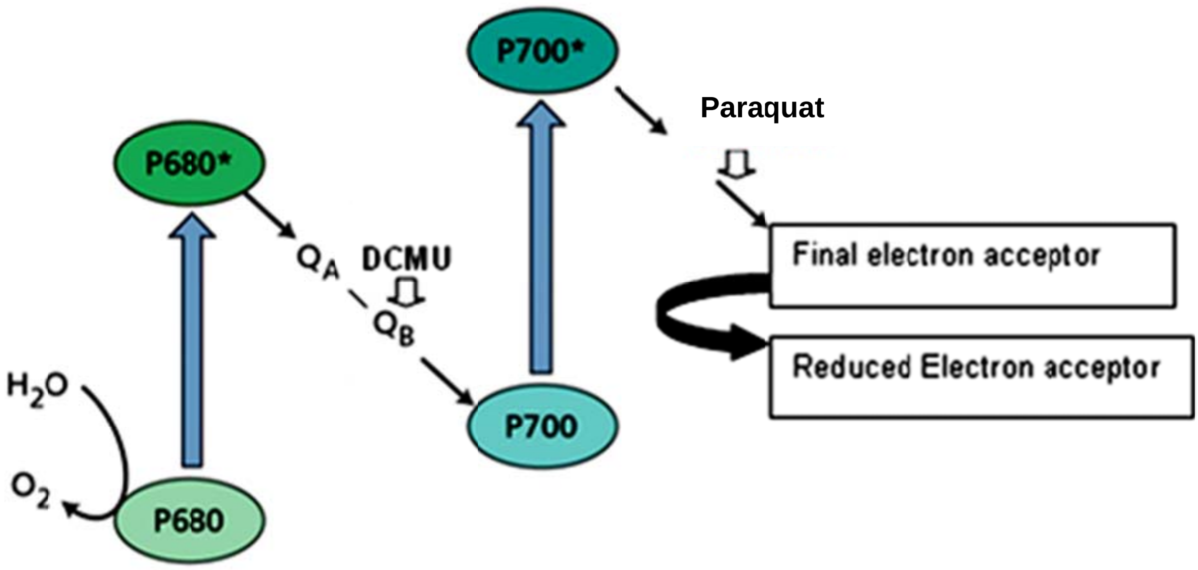
A **I** and **II** only

B **III** and **IV** only

C **I**, **II** and **IV** only

D All of the above

23. The diagram shows the electron transport chain embedded in the thylakoid membrane.

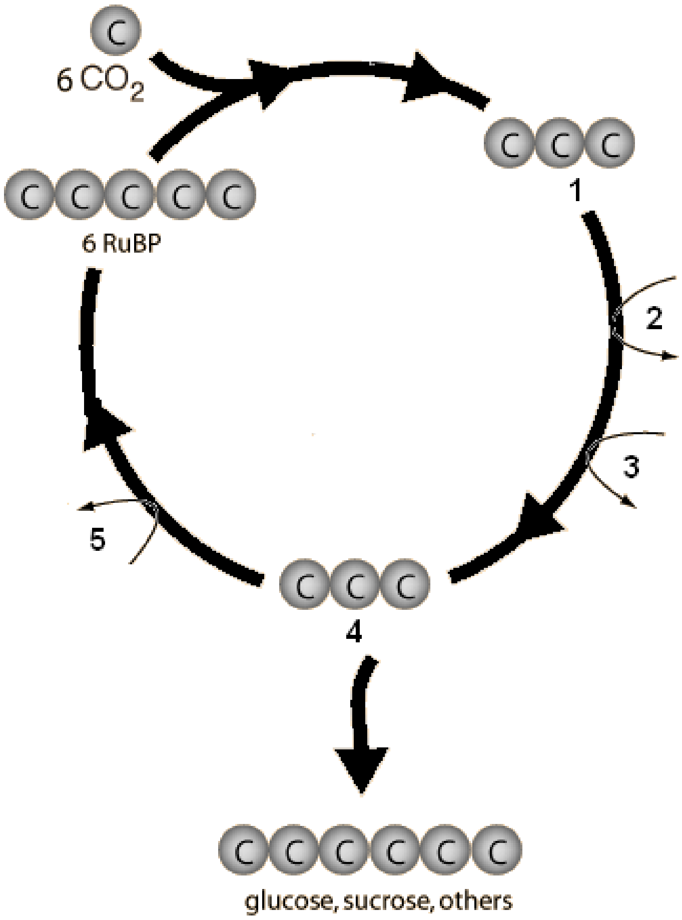


DCMU and paraquat are poisons that disrupt electron transport at the respective positions indicated in the diagram. These two poisons are added separately to isolated chloroplasts. Which of the following ***CORRECTLY*** represents the outcomes in the presence of DCMU and paraquat?

(The sign "+" means produced and the sign "-" means not produced.)

	DCMU			**Paraquat**		
	NADPH	**Oxygen**	**ATP**	**NADPH**	**Oxygen**	**ATP**
A	+	-	+	-	-	+
B	-	+	-	-	+	+
C	-	+	+	+	+	-
D	-	-	-	-	-	+

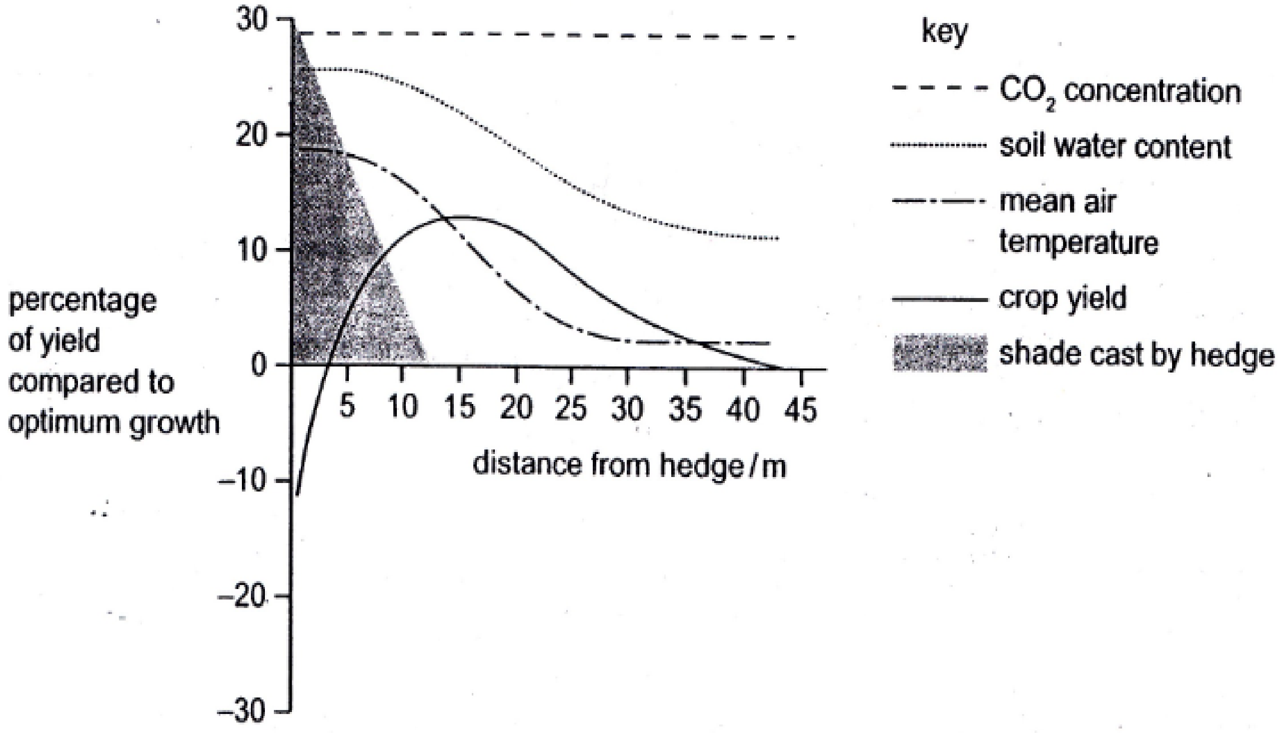
24. The Calvin cycle involves several reactions.



Which row correctly identifies the molecules or reactions at 1 to 5?

	1	2	3	4	5
A	glycerate phosphate	ATP→ADP	NADPH→NADP	triose phosphate	ATP→ADP
B	glycerate phosphate	ATP→ADP	NADH→NAD	triose phosphate	ATP→ADP
C	glycerate phosphate	ADP→ATP	NADP→NADPH	triose phosphate	ADP→ATP
D	triose phosphate	ATP→ADP	NADPH→NADP	glycerate phosphate	ATP→ADP

25. In the diagram, researchers superimposed the changes in four environmental factors on a graph of crop yield against distance from a hedge.



Which factor is most likely to have been the limiting factor in crops 10m from the hedge?

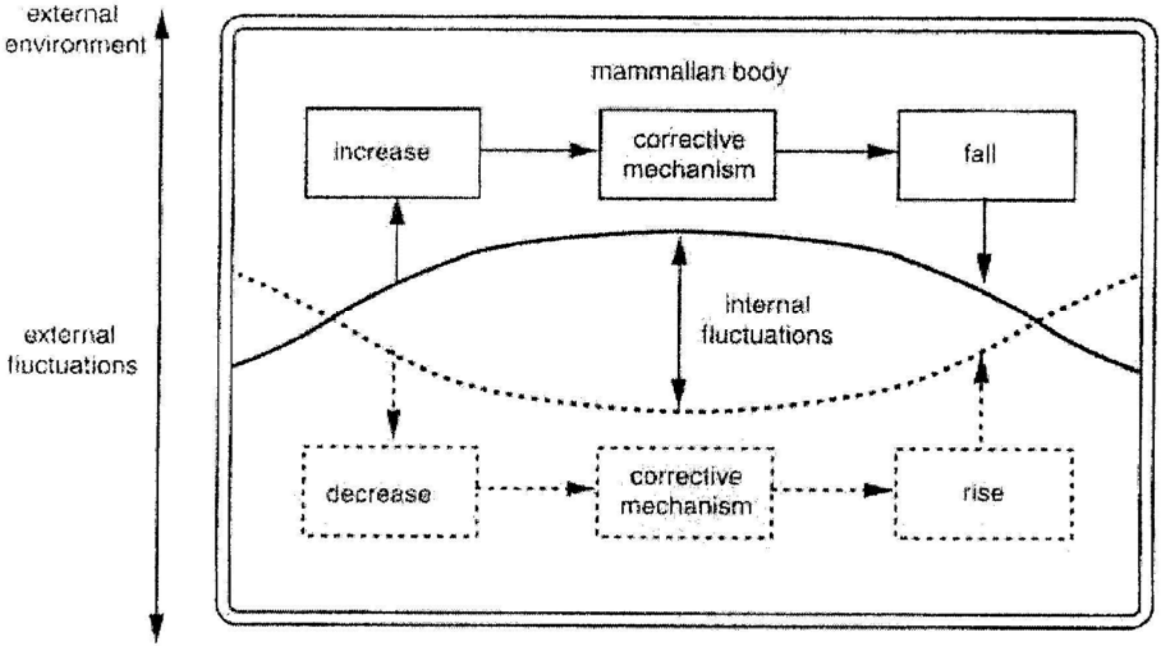
A CO_2 concentration

B Light intensity

C Temperature

D Soil water content

26. The diagram shows some important features of homeostatic mechanisms in the body.



Which of the following ***CORRECTLY*** describes how a mammal maintains a constant internal environment?

A The control center carries out the response to cancel the deviation, causing a fall/rise back to reference point.

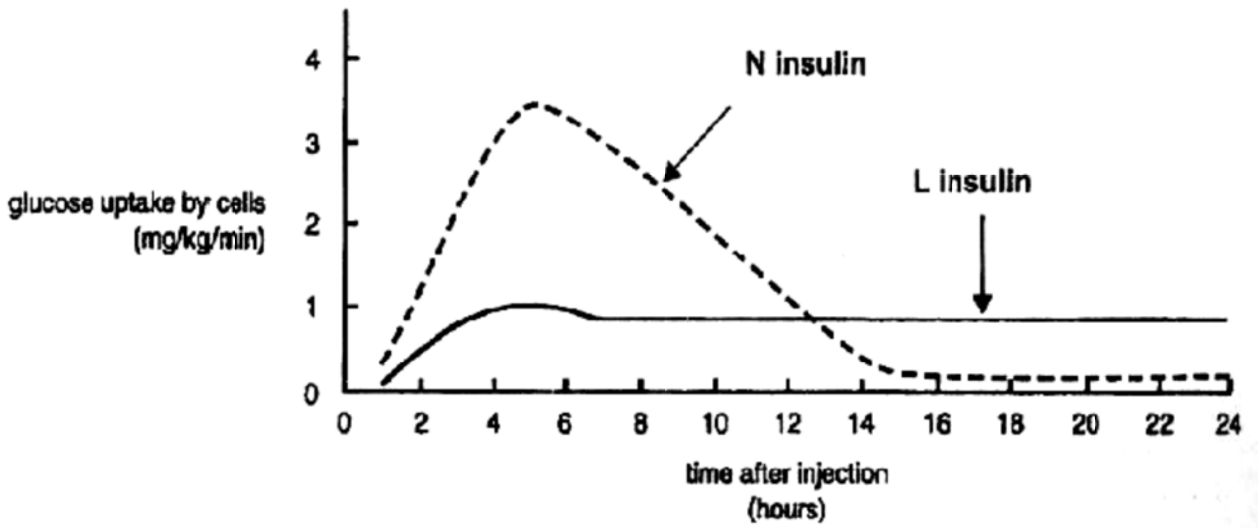
B When there is a deviation from reference point, the detector detects the deviation and transmits the information to the control center.

C The effector receives input from the detector and compares the input with the reference point.

D When the reference point is reached, there is feedback information and the mechanism is triggered to enhance the deviation in the opposite direction from reference point.

27. Insulin is a hormone involved in the regulation of blood glucose levels. Failure to produce insulin results in insulin-dependent (type I) diabetes, and people with this condition must have regular injections of insulin. The effectiveness of two types of insulin was tested. Participants in this test were divided into two groups. One group received N insulin. The second group received L insulin. All participants received the same amount and concentration of the appropriate insulin.

The following graph shows the average results for participants in each of the two groups.



Which of the following statements is/are true about the effectiveness of the two types of insulin?

I **N** insulin needs to be given more frequently than **L** insulin in a day.

II A dose of **N** insulin is more effective than **L** insulin in reducing blood glucose level in one day.

III Using **L** insulin may be more advantageous for a person with type I diabetes.

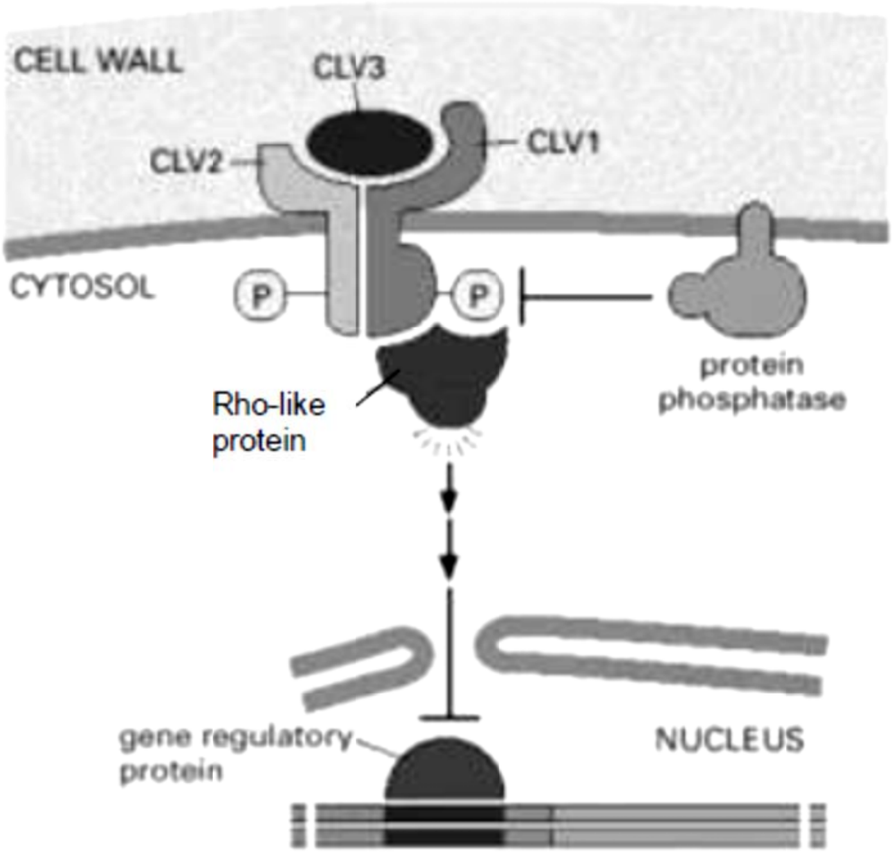
A **I** only

B **II** only

C **II** and **III**

D **I** and **III**

28. The diagram below shows the cell signalling pathway involved in cell proliferation and differentiation in the shoot meristem.



Which of the following statement shows the **CORRECT** sequence of events leading to the activation of the above signalling pathway?

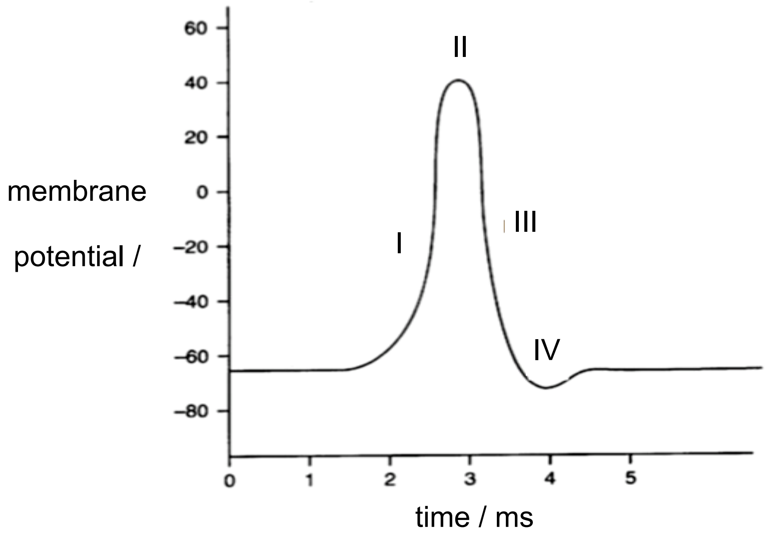
A Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.

B Binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → autophosphorylation of serine/threonine residues → activation of Rho-like protein → activation of gene transcription.

C Binding of CLV3 to subunits CLV2 and CLV1→ dimerisation of CLV2 and CLV1 → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.

D Dimerisation of CLV2 and CLV1 → binding of CLV3 to dimer → dephosphorylation of serine/threonine residues by protein phosphatase → activation of Rho-like protein → activation of gene transcription.

29. The diagram shows a normal action potential.



A purified toxin, charybdotoxin, from the scorpion was tested for its effect on ion channels in nerve cells and was found to block voltage-gated K^+ channels.

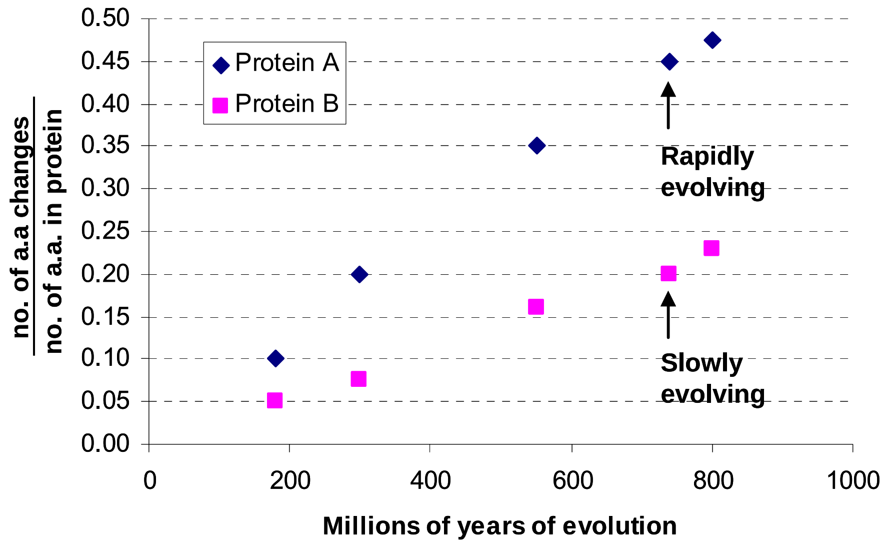
How might the action potential change in a nerve cell affected by a small quantity of charybdotoxin?

	I	II	III	IV
A	increases more slowly	lower	decreases more slowly	does not become hyperpolarised
B	increases more slowly	lower	decreases more slowly	remains hyperpolarised
C	no change	no change	decreases more slowly	does not become hyperpolarised
D	no change	no change	no change	remains hyperpolarised

30. Which combination of definitions is correct?

	Taxonomic group	Phylogeny	Binomial
A	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the names of the genus and species of the organism
B	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the family and genus of the organism
C	a category in a classification system	grouping organisms according to how recently they evolved from a common ancestor	the names of the genus and species of the organism
D	grouping organisms according to how recently they evolved from a common ancestor	survival of the fittest	the name of the species of the organism

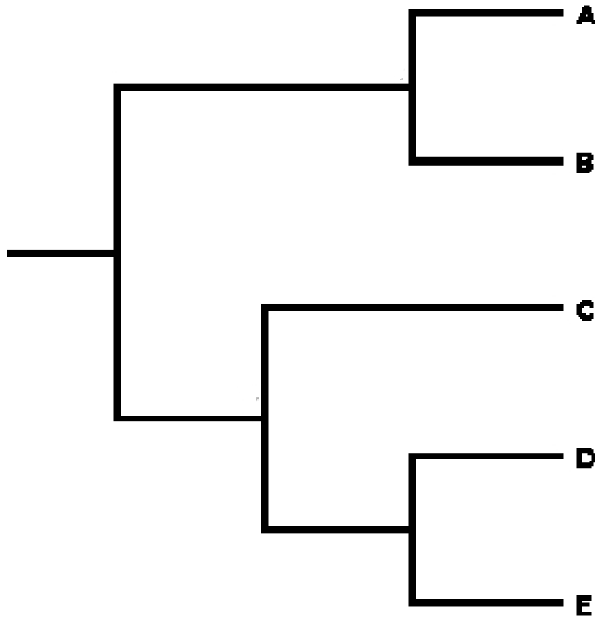
31. The graph below shows the evolution of two different proteins against the evolutionary time that has passed.



From the graph, we can deduce that

A	the difference between the amino acid sequences of protein A and protein B shows how much evolution has happened in the 800 million years.
B	the evolution of protein A is by natural selection while that of protein B is mostly neutral changes that make no difference to how the protein works.
C	protein A has a higher proportion of possible changes that are neutral and hence evolved at a higher rate.
D	protein B has a higher proportion of possible changes that are neutral and hence evolved at a slower rate.

32. The following shows a phylogenetic tree that is based on comparing amino acid sequences.



The amino acid sequence for organism **D** is

Ala – Gly – Leu – Ala – Pro – Ser - Gly - Cys

Which of the following comparisons matches the relationships shown in the phylogenetic tree?

	Organism	Amino acid sequence
A	C	Ala – Pro – Leu – Val - Ala – Gly – Gly – Cys
	E	Cys – Gly – Leu – Val – Pro – Ser – Gly - Ala
B	C	Ala – Pro – Leu – Ala – Pro – Val – Gly - Cys
	E	Ala – Gly – Leu – Val – Pro – Ser - Gly - Ala
C	B	Val - Pro – Leu – Ala – Pro – Ser – Cys - Gly
	C	Ala – Pro – Leu – Val - Ala – Gly – Gly - Cys
D	A	Cys – Gly – Leu – Val – Pro – Ser – Gly - Val
	E	Cys – Gly – Leu – Val – Pro – Ser – Gly - Ala

33. Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?

A high copy number

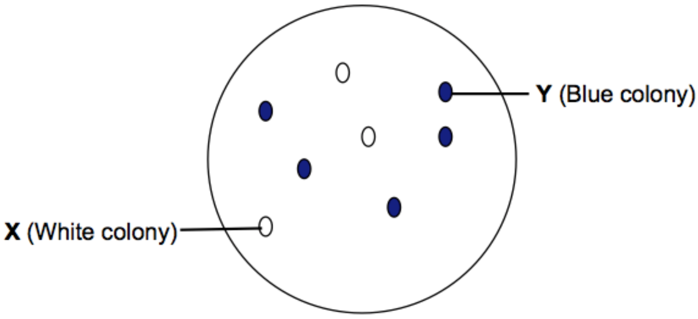
B vulnerable at several sites to a restriction enzyme

C control their own replication

D small in size

34. pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, TetR and lacZ. The multiple cloning site (MCS) of the cloning vector is located within the lacZ gene.

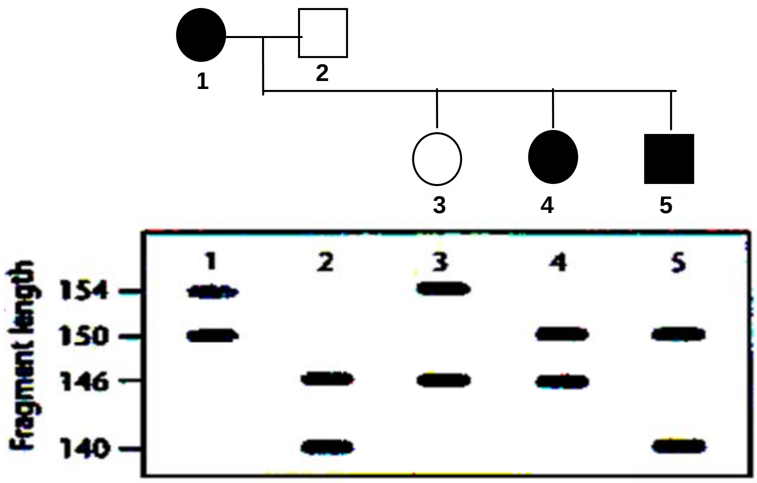
Using SacII as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS of plasmid pGEM-T and the recombinant plasmid is transformed into E. coli cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



Colonies **X** and **Y** are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.21kb and 0.34kb fragments

35. An RFLP genetic marker located on chromosome 13 is used to trace the inheritance of a genetic disease within an affected family. The following pedigree shows two generations of the family with affected individuals represented by filled symbols. Results of the Southern Blot analysis for the corresponding family members are shown in the pedigree.



The allelic variants of the polymorphic RFLP locus are designated by the respective fragment lengths produced in this linkage analysis. Which allele of the RFLP locus is linked to the disease allele?

A Allele **140**

B Allele **146**

C Allele **150**

D Allele **154**

36. Which of the following aims would the creation of a genomic DNA library help to achieve?

A to identify genes that are responsible for liver cell function

B to determine the sequence of telomeric repeats in a person

C to obtain the complete sequence of ras gene

D to study the changing patterns of gene expression during pig development

37. Which uses of the information from the human genome project are generally considered to be unethical?

1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases

2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships

3 cytologists developing tests for only some defective genes

4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes

5 genetic councillors giving specific lifestyle information only to people genetically predisposed to risks

6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes

A 1 and 3

B 1 and 6

C 2 and 6

D 3 and 4

38. Which of the following is not true of adult stem cells during tissue repair?

A The stem cells must have active telomerase.

B The different checkpoints in the cell cycle of the stem cells are activated.

C Mitosis of the stem cells is induced without any stimulus.

D The stem cells will stop dividing after the damaged cells are replaced.

39. In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?

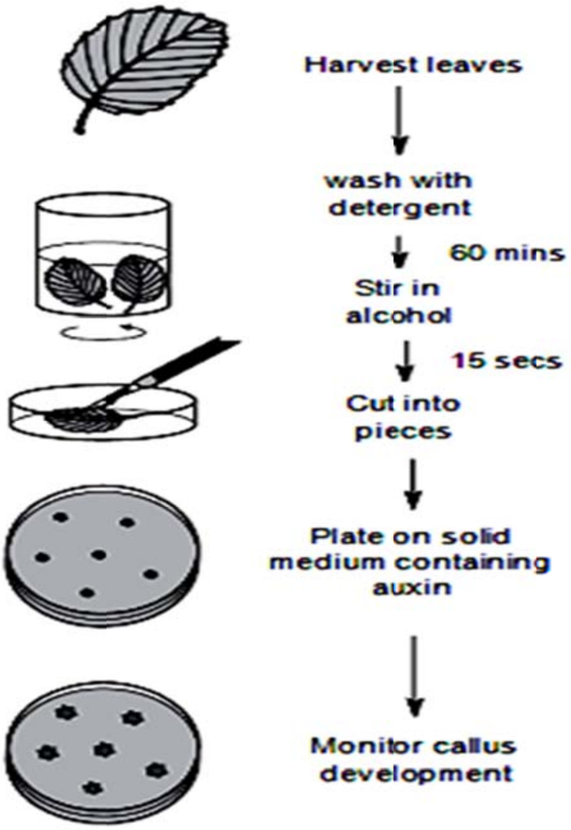
A Chloroplasts are surrounded by a double membrane.

B There are no chloroplasts in pollen of most plant species.

C Chloroplasts are smaller than the nucleus.

D There is no DNA in chloroplasts.

40. Callus tissue is produced using leaves by the technique shown in the diagram below. Using this, a student investigated the effect of two different concentrations of auxin on the volume of callus material produced. The table shows the results obtained over five days.



Time (day)	volume of callus material (cm^3)	
	low concentration of auxin added	high concentration of auxin added
0	1.0	1.0
1	1.3	1.7
2	1.9	4.1
3	3.0	8.3
4	3.9	10.6
5	5.4	12.9

Which of the following statements about the experiment is / are false?

I The leaves are disinfected and sterilised by washing with detergent and alcohol.
II The leaves are wounded in order for the calli to be formed.
III The concentration of auxin affects the volume of callus formed.
IV Continued subculturing at three- to four-week intervals of small cell clusters taken from calli can maintain the callus cultures for long periods of time.

A **III** only
B **I** and **III** only
C **I, II** and **III** only
D None of the above

END OF PAPER